

Effects of intestinal microflora and the environment on the development of asthma and allergy.

The aim of previous research into the causes of allergic diseases, including asthma was mostly to identify potential risk factors in the environment. No major risk factors have been identified, however. Over the past 10 years, focus has, therefore, more been directed towards protective factors that could enhance the development of tolerance to allergens which were previously encountered early in life, but are now lost in modern affluent societies. In particular, the role of childhood infections has been discussed, but so far these studies have not been conclusive. Recent epidemiological studies and experimental research suggest that the microbial environment and exposure to microbial products in infancy modifies immune responses and enhances the development of tolerance to ubiquitous allergens. The intestinal microflora may play a particular role in this respect, as it is the major external driving force in the maturation of the immune system after birth, and animal experiments have shown it to be a prerequisite for normal development of oral tolerance. Recent studies have shown differences in the composition of the microflora between healthy and allergic infants in countries with a high and low prevalence of allergies and between healthy and allergic infants within such countries. These differences are apparent within the first week of life and thus precede clinical symptoms. The use of live microorganisms that might be beneficial to health has a long tradition and the safety is well documented. Very recently, several prospective intervention studies, modifying the gut flora from birth have yielded encouraging results and may suggest a new mode of primary prevention of allergy in the future.